

DYLAN PENTLAND

DPENTLAND@MATH.HARVARD.EDU

[DPENTLAND.GITHUB.IO](https://github.com/dpentland)

EDUCATION

Harvard Ph.D in Mathematics	2022- Present
MIT B.S. in Mathematics	2018 - 2022 5.0/5.0 GPA

PUBLICATIONS

Extensions of mod p representations of local division algebras <i>Journal de Théorie des Nombres de Bordeaux</i>	2024
Filtrations on block subalgebras of restricted universal enveloping algebras <i>Journal of Algebra and Applications</i>	2022
Computing L-polynomials of Picard curves in polylogarithmic time <i>Mathematics of Computation</i>	2022
Coefficients of Gaussian Polynomials Modulo N <i>Electronic Journal of Combinatorics</i>	2020

PAPERS

A prismatic Riemann-Hilbert correspondence

Shows an equivalence of categories between constructible p -Kummer-étale $\mathbf{Z}_p$ -sheaves on the log adic generic fiber $(X, \mathcal{M})_\eta$ for the logarithmic stratification induced by D and certain finitely generated unit Laurent crystals on the log prismatic site. We verify p^+ -perverse t -exactness, give a local description of finitely generated unit crystals on a framed log affine in terms of (φ, Γ) -modules, and show pushforward compatibility for strict log smooth proper morphisms. Can be accessed [here](#).

Syntomification and crystalline local systems

Extends a result of Bhatt-Lurie showing there is an equivalence of categories between reflexive sheaves on the stack $\mathbf{Z}_p^{\text{Syn}}$ and $\mathbf{Z}_p$ -lattices in crystalline representations to the general case of smooth $X/\text{Spf } \mathcal{O}_K$, where $K/\mathbf{Q}_p$ is a finite extension. I show $\text{Perf}(X^{\text{Syn}})[1/p]$ is a certain derived category of admissible filtered F -isocrystals when X is smooth proper. The key input is a calculation of syntomic cohomology with coefficients over $\mathcal{O}_K$ via an explicit syntomic complex. Submitted to *Algebra and Number Theory*. Can be accessed [here](#).

Extensions of mod p representations of local division algebras

Determines the extension groups of smooth irreducible representations of a division algebra D over a non-Archimedean local field by studying the structure of $H^\bullet(I_1, \pi)$ where $I_1 = 1 + \varpi_D \mathcal{O}_D$ and π is some irreducible representation of $D^\times$. Published in *Journal de Théorie des Nombres de Bordeaux*. Can be accessed [here](#).

Filtrations on block subalgebras of restricted universal enveloping algebras

Studies the associated graded algebras for the PBW filtration and related filtrations on blocks of restricted universal enveloping algebras. Published in the *Journal of Algebra and its Applications*. Can be accessed [here](#).

Computing L -polynomials of Picard curves in polylogarithmic time

Develops and implements a practical algorithm to compute the zeta function of a curve in genus > 2 . The implementation was done in SAGE and PARI/GP, and has applications to cryptography. Published in *Mathematics of Computation*. Can be accessed [here](#).

Coefficients of Gaussian Polynomials Modulo N

Resolves and extends a conjecture of Prof. Richard Stanley on periods of coefficients in q -binomial coefficients modulo N . Published in *The Electronic Journal of Combinatorics*. Can be accessed [here](#).

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AWARDS

NSF Fellowship	2022
Regeneron STS Finalist	2018

TALKS

A prismatic Riemann-Hilbert correspondence	2026
Boston University Number Theory Seminar	
A prismatic Riemann-Hilbert correspondence	2026
Ramanujan-Serre seminar	
Syntomification and crystalline local systems	2025
Harvard Number Theorists' seminar	

TEACHING AND OTHER ACTIVITIES

Linear Algebra	2025 Spring
Taught Math 21b at Harvard.	
Summer Tutorial	2023 Summer
Taught a Harvard summer tutorial on Bass-Serre theory.	
DRP Mentor	2023-
Mentored an undergraduate students in étale cohomology, p -adic Hodge theory, rational points on elliptic curves, group theory, analytic number theory.	
Head Counselor at PROMYS	2022
Managed program for motivated high school students learning number theory.	